

$\mathbb{R}^4$

$M_0^3 \subset \mathbb{R}^4$

Ilmanen

 M_0

sheet

$1 \leq q \leq \lfloor \lambda(M_0) \rfloor$

$V_{-1}^\infty = m|\Sigma|$ $mF_{0,1}(\Sigma) = \Theta(x_0, T) \leq \lambda(M_0)$ $m \leq \lfloor \lambda(M_0)/F_{0,1}(\Sigma) \rfloor$

2026 8 11

Danus verifier

$M_0^3 \subset \mathbb{R}^4$	
$M^3 \subset \mathbb{R}^4$	
	$q \leq \lfloor \lambda(M_0) \rfloor$
M_0	1
	Brakke

$\lambda(M_0) < 2$ 1 α -Andrews

Type-I

sheet

1.

			$M_s^{(i)} = lambda_i(M_{T+\lambda_i^{-2}s} - x_0), \quad s < 0,$		
Brakke			$\mathcal{T} = \{V_s^\infty\}_{s<0}$		
			$\theta^3(\ V_s^\infty\ ,x) = 1$		
			$\ V_s^\infty\ $		
Haslhofer	2025	5.2	Conjecture 5.10		blow-up limit
		Bamler-Kleiner	$\mathbb{R}^3$		
		$M^3 \subset \mathbb{R}^4$			
Bamler-Kleiner		$\mathbb{R}^3$	2	3	Li-Wang Type-I
		$\mathbb{R}^3$			

2.

2.1

varifold V	Gaussian F -
$F_{y,\tau}(V) = (4\pi\tau)^{-3/2} \int e^{- x-y ^2/(4\tau)} d\ V\ (x), \qquad \tau > 0.$	
M	
$\lambda(M) = \sup_{y \in \mathbb{R}^4, \tau > 0} F_{y,\tau}(M).$	
Huisken	
$\lambda(M_t) \leq \lambda(M_0), \qquad 0 \leq t < T. \tag{2.1}$	

2.2 **sheet**

2.1	V_s^∞	x						
			$V_s^\infty = q \Gamma $					
Γ	$q \in \mathbb{N}$							
	$1 \leq q \leq \lfloor \lambda(M_0) \rfloor.$							

(2.2)

$$(2.1) \qquad \qquad \qquad \lambda(M_0) \qquad \qquad \text{varifold} \qquad \qquad \text{Gaussian}$$

$$\lambda(V_s^\infty) \leq \lambda(M_0). \tag{2.3}$$

$$x \qquad \qquad x \qquad \qquad \tau \downarrow 0 \quad \text{Gaussian} \qquad \qquad \Gamma \qquad x$$

$$\lim_{\tau \downarrow 0} F_{x,\tau}(V_s^\infty) = q.$$

$$q \leq \lambda(V_s^\infty) \leq \lambda(M_0).$$

$$q \qquad \qquad (2.2)$$

$$M_0 \qquad \qquad \qquad \qquad \qquad \qquad \qquad \qquad \qquad \qquad \qquad \qquad \qquad C_0^\alpha$$

$$\lambda(M_0) \qquad \qquad \lambda(C_0^\alpha) \qquad \qquad \qquad \qquad \qquad \qquad \qquad \qquad \qquad \qquad \qquad \qquad \qquad 2$$

2.3 self-shrinker

2.2

$$\Sigma^3 \subset \mathbb{R}^4 \qquad \qquad \qquad V_{-1}^\infty = m|\Sigma|, \qquad m \in \mathbb{N},$$

$$\qquad \qquad \qquad \qquad \qquad \qquad \qquad \qquad \qquad \qquad \qquad \text{self-shrinker}$$

$$mF_{0,1}(\Sigma) = \Theta(x_0,T) \leq \lambda(M_0), \tag{2.4}$$

$$m \leq \left\lfloor \frac{\lambda(M_0)}{F_{0,1}(\Sigma)} \right\rfloor \leq \lfloor \lambda(M_0) \rfloor. \tag{2.5}$$

$$\text{Huisken} \qquad \qquad (x_0,T) \qquad \qquad \text{Gaussian density} \qquad \qquad \qquad \text{Gaussian}$$

$$\begin{aligned} \Theta(x_0,T) &= F_{0,1}(V_{-1}^\infty) \\ &= mF_{0,1}(|\Sigma|). \end{aligned}$$

$$(x_0,T)$$

$$\Theta(x_0,T) \leq \lambda(M_0).$$

$$\text{self-shrinker} \qquad \qquad \qquad \qquad \qquad \qquad \qquad \qquad \qquad \qquad \qquad \text{Gaussian} \qquad \qquad \qquad 1$$

$$F_{0,1}(\Sigma) = \lambda(\Sigma) \geq 1.$$

$$(2.5)$$

$$\textbf{2.3} \qquad \lambda(M_0) < 2 \qquad \qquad \qquad V_{-1}^\infty = m|\Sigma| \qquad \qquad m = 1$$

$$\lambda(M_0) \leq 2 \qquad m \leq 2$$

 Σ

$$\Lambda_3 = \lambda(S_{\sqrt{6}}^3) = \frac{3}{2}\sqrt{6\pi}e^{-3/2} \approx 1.45265.$$

2.4 shrinker

2.2 Σ

$$m \leq \left\lfloor \frac{\lambda(M_0)}{\Lambda_3} \right\rfloor. \quad (2.6)$$

$$\lambda(M_0) < 2\Lambda_3 \approx 2.9053 \quad m = 1$$

$\Sigma^3 \subset \mathbb{R}^4$ Jordan-Brouwer

$$H_3(\Sigma; \mathbb{Z}) \cong \mathbb{Z}.$$

Hershkovits-White sharp entropy bound

$$F_{0,1}(\Sigma) = \lambda(\Sigma) \geq \lambda(S_{\sqrt{6}}^3) = \Lambda_3.$$

(2.4) (2.6)

3. “

$$m \leq \left\lfloor \frac{\lambda(M_0)}{F_{0,1}(\Sigma)} \right\rfloor,$$

sheet $q \leq \lfloor \lambda(M_0) \rfloor$

$$\Sigma \quad 2.4$$

$$m \leq \lfloor \lambda(M_0)/\Lambda_3 \rfloor.$$

M_0	universal bound
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universal bound	1
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1.

2.

3.

$$m \qquad S^3$$

S^3

varifold

$m|S^3|$

Gaussian

$$mF(S^3) \quad m \rightarrow \infty$$

4.

$P \subset \mathbb{R}^4$				
	$P_i^+ = P + \varepsilon_i \nu, \qquad P_i^- = P - \varepsilon_i \nu, \qquad \varepsilon_i \downarrow 0.$			
$P_i^+ \cup P_i^-$	sheet			
	$ P_i^+ + P_i^- \longrightarrow 2 P $			
integral varifold	sheet			
•	P			
• m	P	m	1	
• degree-one				1
Type-I				
	$ A \leq \frac{C_A}{\sqrt{T-t}}$ (4.1)			
	$ A_{M_s^{(i)}} \leq \frac{C_A}{\sqrt{-s}},$ (4.2)			
$s = -1$				varifold
sheet	(4.2)	sheet	1	
“	density	1”	varifold	
	$V = m P ,$			
	$\Theta(V,0) = m.$			

5.

Danus	“	sheets	”
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$$x\in M_t\qquad \sigma_M(x,t)\qquad x\qquad +\infty$$

$$\rho_{(x_0,T)}(x,t) = (4\pi(T-t))^{-3/2}e^{-|x-x_0|^2/(4(T-t))},$$

$$a_-:=\min\{a,0\}$$

$$\text{5.1 Type-I} \qquad (4.1) \qquad C<\infty \qquad t<T$$

$$\int_{M_t}\left(\log\frac{\sigma_M(x,t)}{\sqrt{T-t}}\right)_-\rho_{(x_0,T)}(x,t)\,d\mathcal{H}^3(x)\geq -C. \tag{5.1}$$

$$t_i=T-\lambda_i^{-2}$$

$$\lambda_i(M_{t_i}-x_0)\longrightarrow m|\Sigma|$$

$$\text{varifold} \qquad \Sigma \qquad m \qquad m=1$$

$$(4.2) \qquad s=-1 \qquad \text{varifold}$$

$$m$$

$$m\geq 2 \qquad \Sigma \qquad \text{Gaussian} \qquad K \qquad K \qquad d_i$$

$$\lambda_i\sigma_M(x,t_i)\leq c, d_i\longrightarrow 0.$$

$$\sqrt{T-t_i}=\lambda_i^{-1}$$

$$\frac{\sigma_M(x,t_i)}{\sqrt{T-t_i}}=\lambda_i\sigma_M(x,t_i)\longrightarrow 0.$$

$$-\infty \qquad \text{Gaussian} \qquad \text{Fatou}$$

$$(5.1) \qquad t_i \qquad -\infty \qquad m=1$$

$$M_0^3\subset \mathbb{R}^4 \qquad (5.1)$$

$$\text{sheet-separation}$$

$$\mathbf{6.}$$

$$\mathbf{6.1}$$

$$2.3$$

$$\lambda(M_0)<2\quad\Longrightarrow\quad\text{ooooooooo}1.$$

6.2 α -Andrews

White Haslhofer-Kleiner

6.3

Chodosh-Mantoulidis-Schulze $\mathbb{R}^4$ 2
generic singularities generic flow

6.4 current

currents Σ current varifold m
current

$\lambda(M_0) < 3$ $\iff$ current $\iff m = 1.$

current 1 3

6.5 shrinker

rescaled time τ self-
shrinker Σ_c C^4 $u_{\text{gap}} > 0$
rescaled MCF

$\partial_\tau u_{\text{gap}} = L_{\Sigma_c} u_{\text{gap}} + P^{ij} \nabla_i \nabla_j u_{\text{gap}} + B^i \nabla_i u_{\text{gap}} + C u_{\text{gap}},$

$\tau \rightarrow \infty$

$L_{\Sigma_c} = \Delta - \frac{1}{2} \langle x, \nabla \cdot \rangle + |A|^2 + \frac{1}{2}.$

gap Gaussian 1/2 Gaussian C^4
“ ”

7. “ ”

$M_0^3 \subset \mathbb{R}^4$ $m \geq 2$

$m S^3_{\sqrt{-6s}} $	m	self-similar integral Brakke flow	
$m \quad S^3$	m		
m	m	Brakke	
Chen-Sun $\mathbb{R}$	2	$M^3 \subset \mathbb{R}^4$	
ancient pancake blow-down	2	ancient flow	$t \rightarrow -\infty$ blow-down tangent flow

$$0 < R_1 < R_2 < \cdots < R_m$$

$$r_k(t)^2=R_k^2-6t.$$

$$T=R_1^2/6\qquad (0,T)$$

sheet

8.

8.1

$$n \qquad M \mapsto \lambda M$$

$$\int_{\lambda M} |A|^p \, d\mu = \lambda^{n-p} \int_M |A|^p \, d\mu.$$

$$\begin{array}{ll} p=2 & p=3 \text{ Bamler-Kleiner} \\ n=3 & L^3 \end{array}$$

8.2

$(\partial_t - \Delta) \log \sigma \geq 0$

5.1

Gaussian

8.3 Jacobi

sheet-gap shrinker Gaussian L - “ L - shrinker” shrinker Jacobi “ ” self- Gaussian

8.4 varifold current

varifold sheet current sheets 2 $m = 1$

8.5 Type-I

$|A|$ sheet “ m ” sheet C^∞ sheet

9.

1.

Ilmanen

$M^3 \subset \mathbb{R}^4$

$\mathbb{R}^3$
2.

sheet

$q \leq \lfloor \lambda(M_0) \rfloor$

$V_{-1} = m|\Sigma|$

(2.5)
3.

Brakke

ancient blow-down

tangent flow
4.

sheets

unit-weight varifold
5.

(5.1)

Gaussian

Type-I
- $m = 1$

L^3

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